

GENERAL CHEMISTRY



Chapter 16 Acid-Base Equilibria

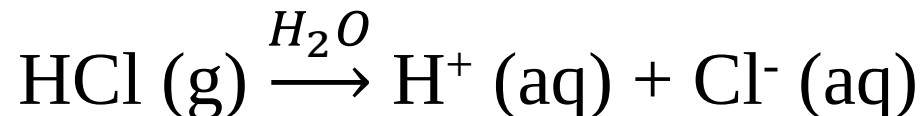
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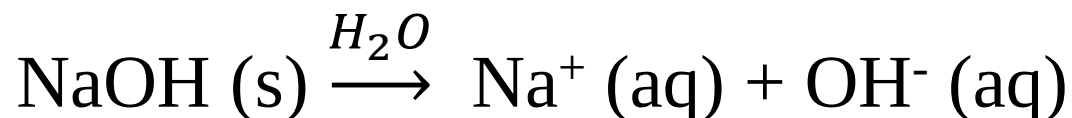
16-1 Acids and Bases: A Brief Review

◆ Arrhenius

- An **acid** is a substance that, when dissolved in water, increases the concentration of **hydrogen** ions, **H⁺**.



- A **base** is a substance that, when dissolved in water, increases the concentration of **hydroxide** ions, **OH⁻**.



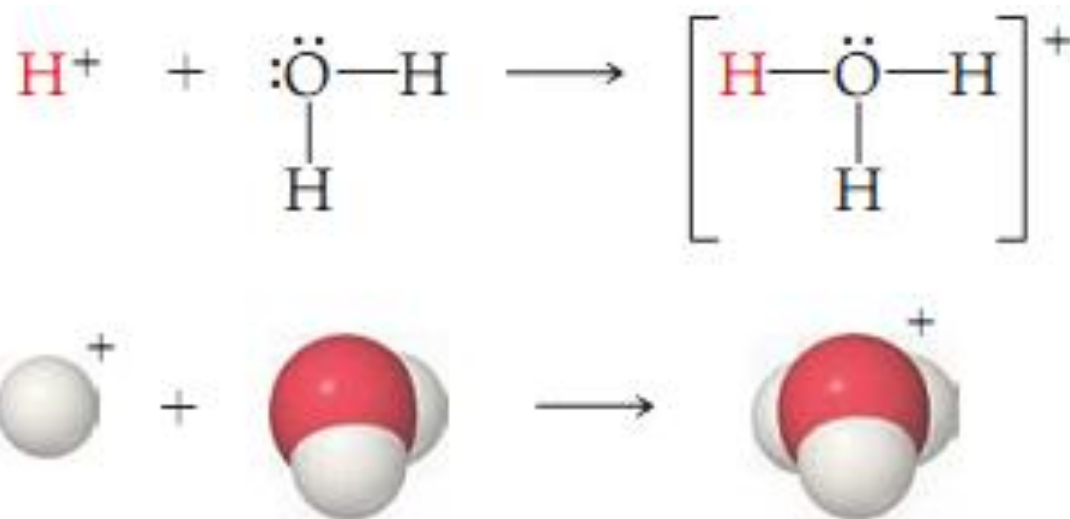
16-2 Brønsted–Lowry Acids and Bases

- ◆ **Brønsted–Lowry** : *Acid–base reactions involve the transfer of H^+ ions from one substance to another*
 - An acid is a proton donor.
 - ... must have a removable proton
 - A base is a proton acceptor.
 - ...must have a pair of nonbonding electrons

The H^+ Ion in Water



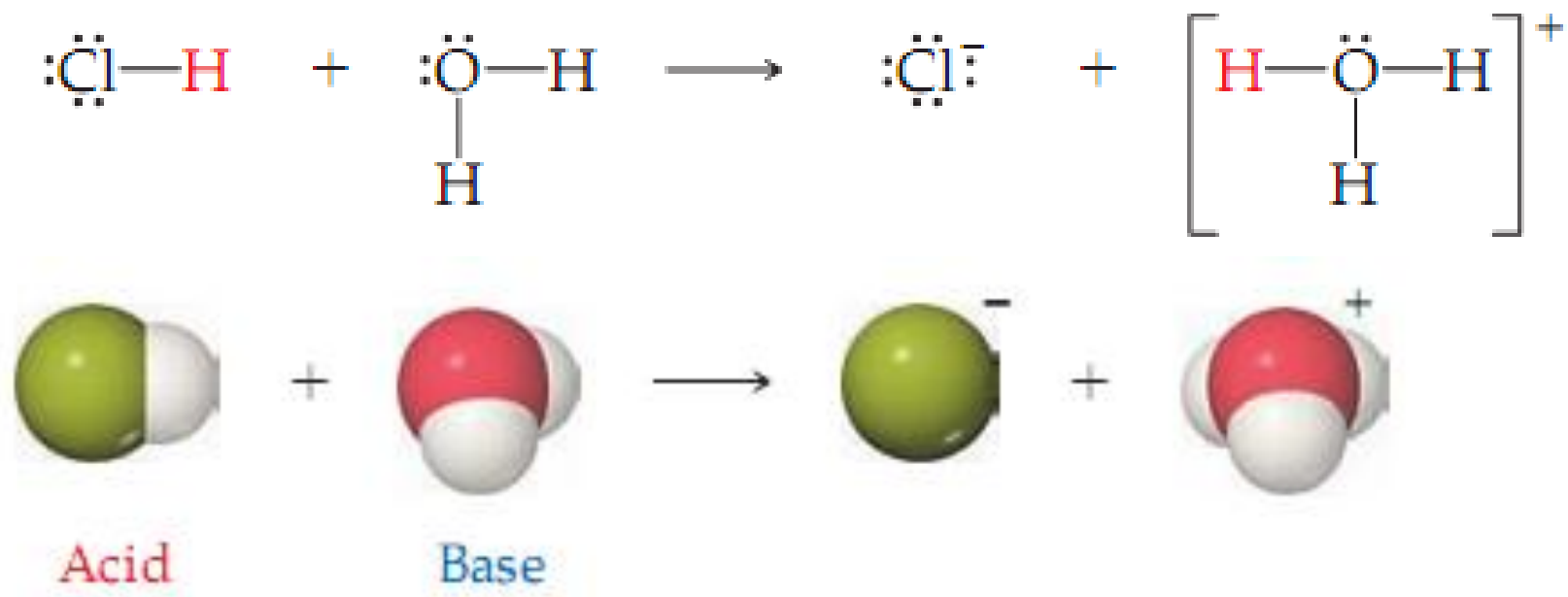
The H^+ ion interacts strongly with any source of electron density, such as the nonbonding electron pairs on the oxygen atoms of water molecules



Chemists use the notations H^+ and H_3O^+ interchangeably

Proton-Transfer Reactions

What Happens When an Acid Dissolves in Water?

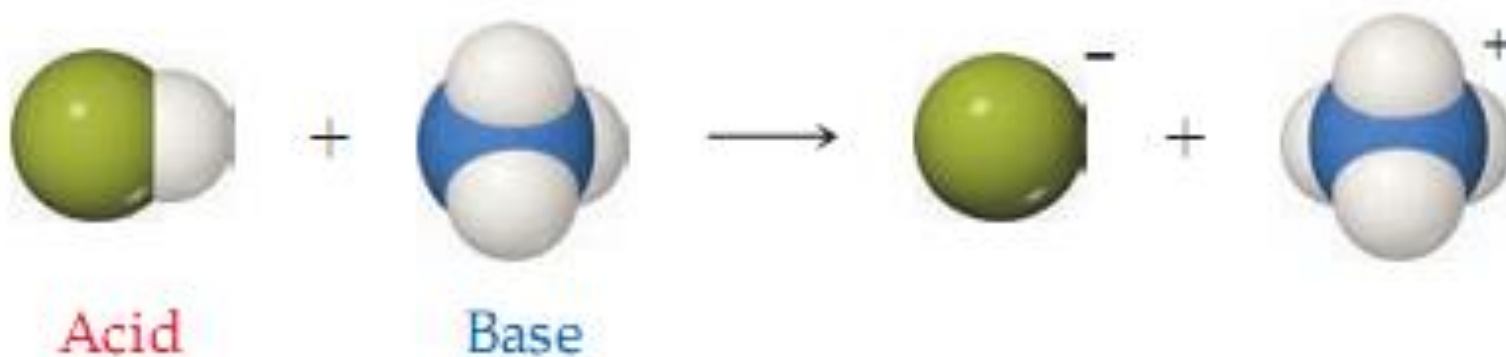
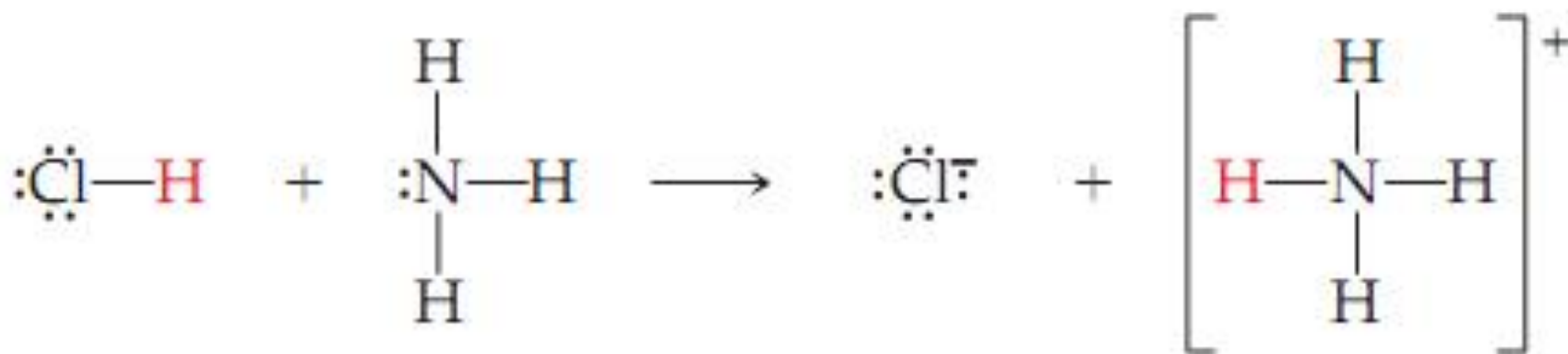


proton donor proton acceptor conjugate base hydronium ion

- ◆ Water acts as a Brønsted–Lowry base and abstracts a proton from the acid.
- ◆ The **conjugate base** of the acid and a **hydronium ion** are formed.

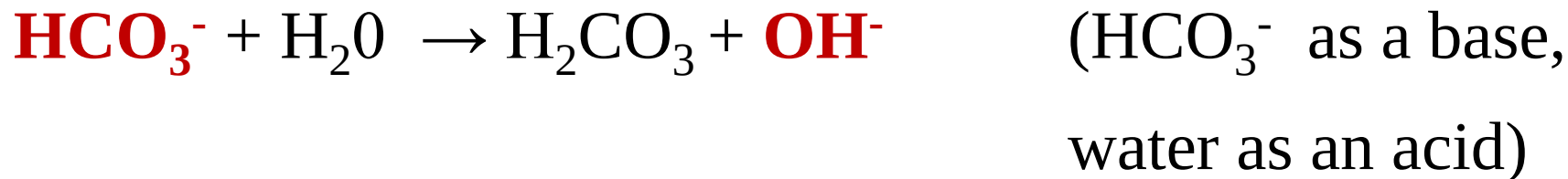
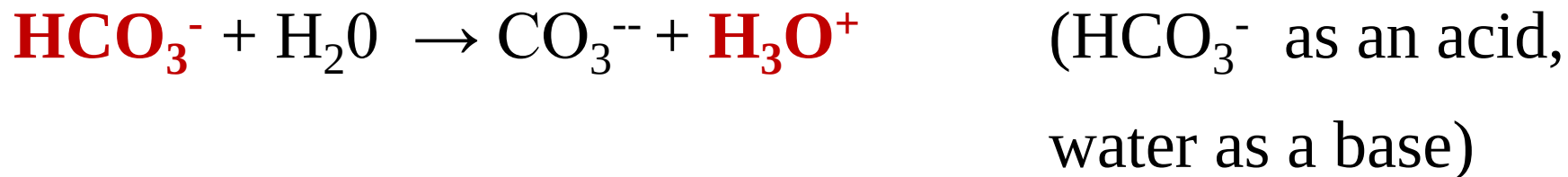
Proton-Transfer Reactions

The Brønsted–Lowry concept also applies to reactions that do not occur in aqueous solutions



Proton-Transfer Reactions

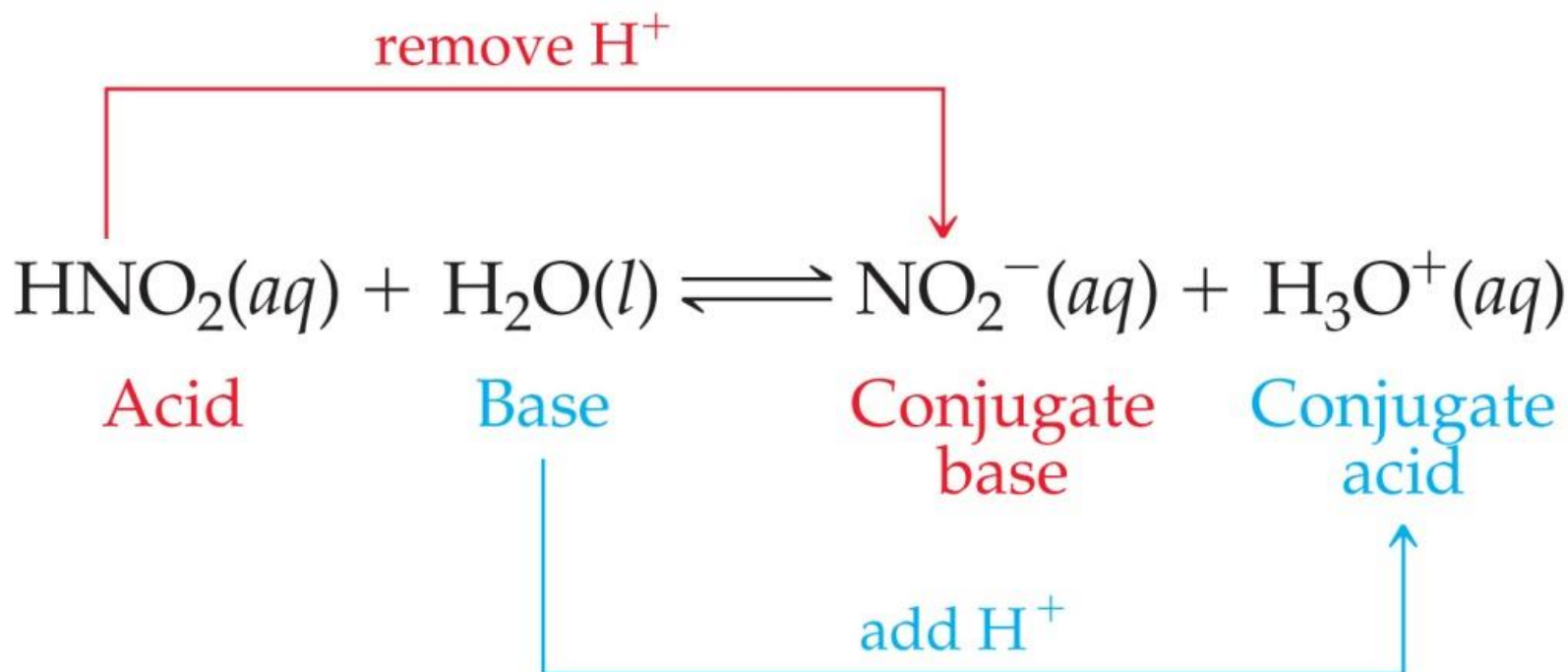
H₂O and HCO₃⁻ can be either acid or base



They are **amphiprotic**

Conjugate Acids and Bases

- ◆ The term **conjugate** comes from the Latin word “conjugare,” meaning “to join together.”
- ◆ Reactions between acids and bases always yield their conjugate bases and acids.



EXAMPLE

(a) What is the conjugate base of



(b) What is the conjugate acid of



EXAMPLE

The hydrogen sulfite ion (HSO_3^-) is amphoteric. Write an equation for the reaction of HSO_3^- with water

(a) in which the ion acts as an acid



acid

conjugate base

(b) in which the ion acts as a base.



base

conjugate acid

In both cases identify the conjugate acid–base pairs.

Relative Strengths of Acids and Bases

ACID		BASE	
Strong acids	HCl	Cl ⁻	Negligible basicity
	H ₂ SO ₄	HSO ₄ ⁻	
	HNO ₃	NO ₃ ⁻	
	H ₃ O ⁺ (aq)	H ₂ O	
	HSO ₄ ⁻	SO ₄ ²⁻	Weak bases
	H ₃ PO ₄	H ₂ PO ₄ ⁻	
	HF	F ⁻	
	CH ₃ COOH	CH ₃ COO ⁻	
	H ₂ CO ₃	HCO ₃ ⁻	
	H ₂ S	HS ⁻	
Weak acids	H ₂ PO ₄ ⁻	HPO ₄ ²⁻	Conjugate bases of substances with negligible acidity are exceedingly strong
	NH ₄ ⁺	NH ₃	
	HCO ₃ ⁻	CO ₃ ²⁻	
	HPO ₄ ²⁻	PO ₄ ³⁻	
	H ₂ O	OH ⁻	
Negligible acidity	OH ⁻	O ²⁻	Strong bases
	H ₂	H ⁻	
	CH ₄	CH ₃ ⁻	

Strong acids are completely dissociated in water

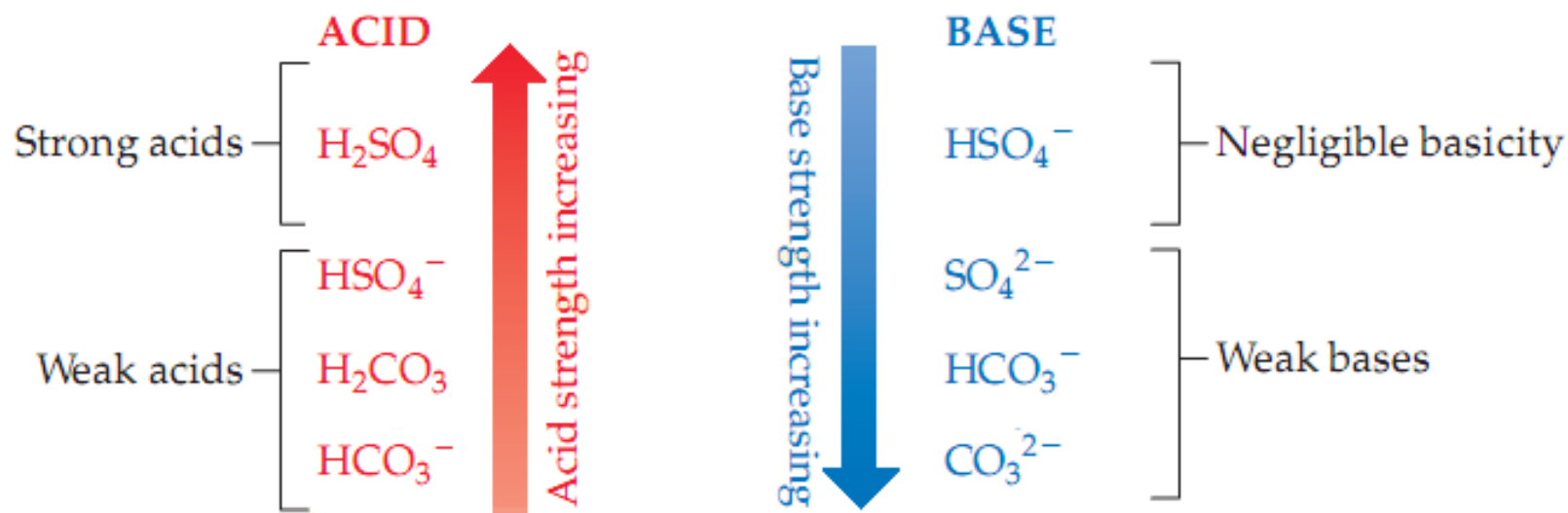
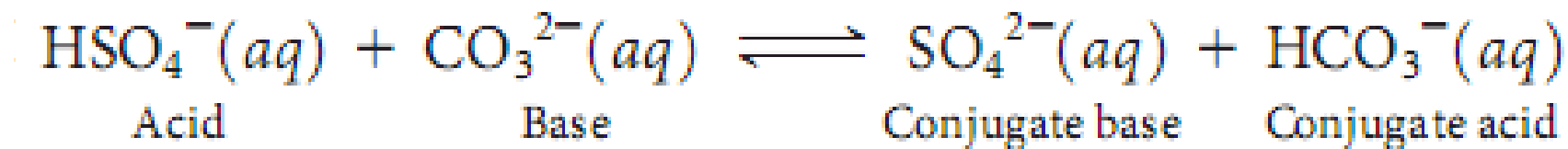
(The conjugate base of a strong acid shows negligible basicity)

Substances with negligible acidity do not dissociate in water

Conjugate bases of substances with negligible acidity are exceedingly strong

EXAMPLE

Predict whether the equilibrium of the following proton-transfer reaction lies to the left ($K_c < 1$) or to the right ($K_c > 1$)?



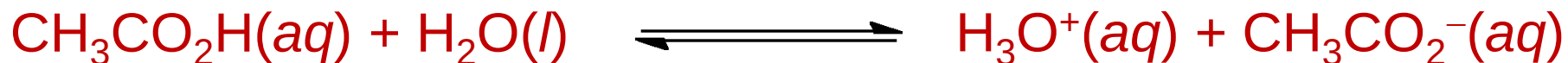
CO_3^{2-} is the stronger base, therefore it will get the proton preferentially
➔ The reaction lies to the right ($K_c > 1$)?

Acid and Base Strength

- ◆ In any acid–base reaction, the equilibrium will favor the reaction that moves the proton to the stronger base to form the weaker acid and the weaker base.



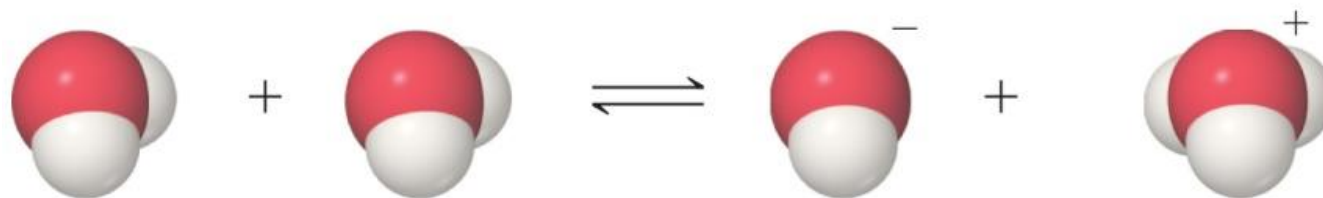
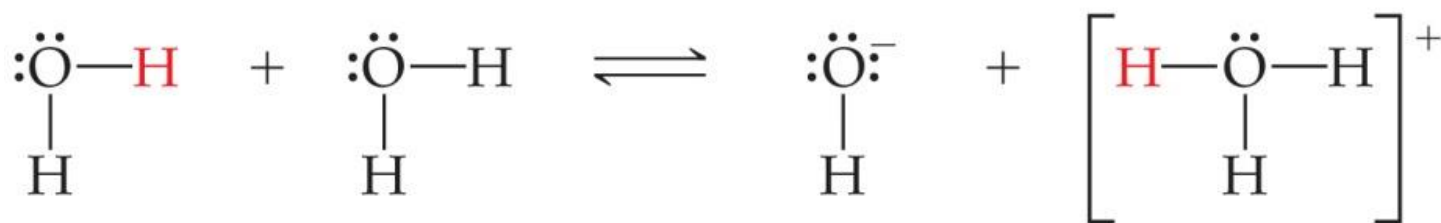
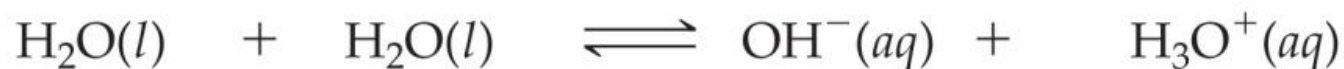
H_2O is a much stronger base than Cl^- , so the equilibrium lies **completely to the right**



Acetate is a stronger base than H_2O , so **the reverse reaction is favored** more than the forward reaction

16-3 The Autoionization of Water

- ◆ As we have seen, water is amphoteric.
- ◆ In pure water, a few molecules act as bases and a few act as acids.
- ◆ This is referred to as **autoionization**.



Acid

Base

The Ion Product of Water

- ◆ The equilibrium expression for the autoionization of water is

$$K_c = [\text{H}_3\text{O}^+] [\text{OH}^-]$$

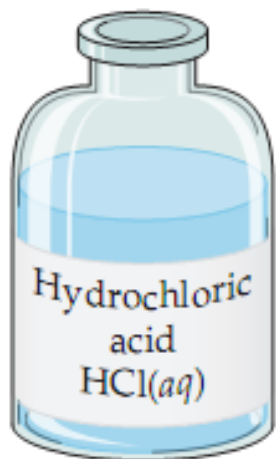
- ◆ This special equilibrium constant is referred to as the **ion product constant** for water, K_w .
- ◆ At 25°C, $K_w = 1.0 \times 10^{-14}$

$$[\text{H}^+][\text{OH}^-] = (x)(x) = 1.0 \times 10^{-14}$$

$$x^2 = 1.0 \times 10^{-14}$$

$$x = 1.0 \times 10^{-7} \text{M} = [\text{H}^+] = [\text{OH}^-]$$

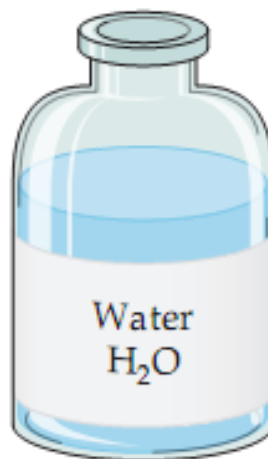
The relative concentrations of H^+ and OH^- in aqueous solutions at 25 °C



Acidic solution

$$[\text{H}^+] > [\text{OH}^-]$$

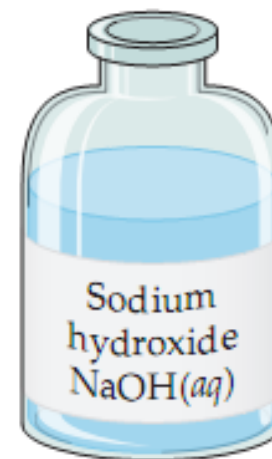
$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$



Neutral solution

$$[\text{H}^+] = [\text{OH}^-]$$

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$



Basic solution

$$[\text{H}^+] < [\text{OH}^-]$$

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$

EXAMPLE

Calculate the concentration of $\text{H}^+(\text{aq})$ in

(a) a solution in which $[\text{OH}^-]$ is 0.010 M,

(b) a solution in which $[\text{OH}^-]$ is 1.8×10^{-9}

The temperature of these solutions is 25°C

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$

$$[\text{H}^+] = \frac{(1.0 \times 10^{-14})}{[\text{OH}^-]}$$

$$\text{a) } [\text{H}^+] = \frac{(1.0 \times 10^{-14})}{[\text{OH}^-]} = \frac{1.0 \times 10^{-14}}{0.010} = 1.0 \times 10^{-12} \text{ M}$$

$$\text{b) } [\text{H}^+] = \frac{(1.0 \times 10^{-14})}{[\text{OH}^-]} = \frac{1.0 \times 10^{-14}}{1.8 \times 10^{-9}} = 5.6 \times 10^{-6} \text{ M}$$

16-4 The pH scale

pH is the negative logarithm in base 10 of the concentration of hydronium ion:

$$\text{pH} = -\log [\text{H}_3\text{O}^+]$$

◆ **For pure water at 25°C,**

$$K_w = [\text{H}_3\text{O}^+] [\text{OH}^-] = 1.0 \times 10^{-14}$$

and

$$[\text{H}_3\text{O}^+] = [\text{OH}^-]$$

Thus,

$$[\text{H}_3\text{O}^+] = \sqrt{1.0 \times 10^{-14}} = 1.0 \times 10^{-7}$$

$$\text{pH} = -\log (1.0 \times 10^{-7}) = 7.00$$

pH of Acids and Bases

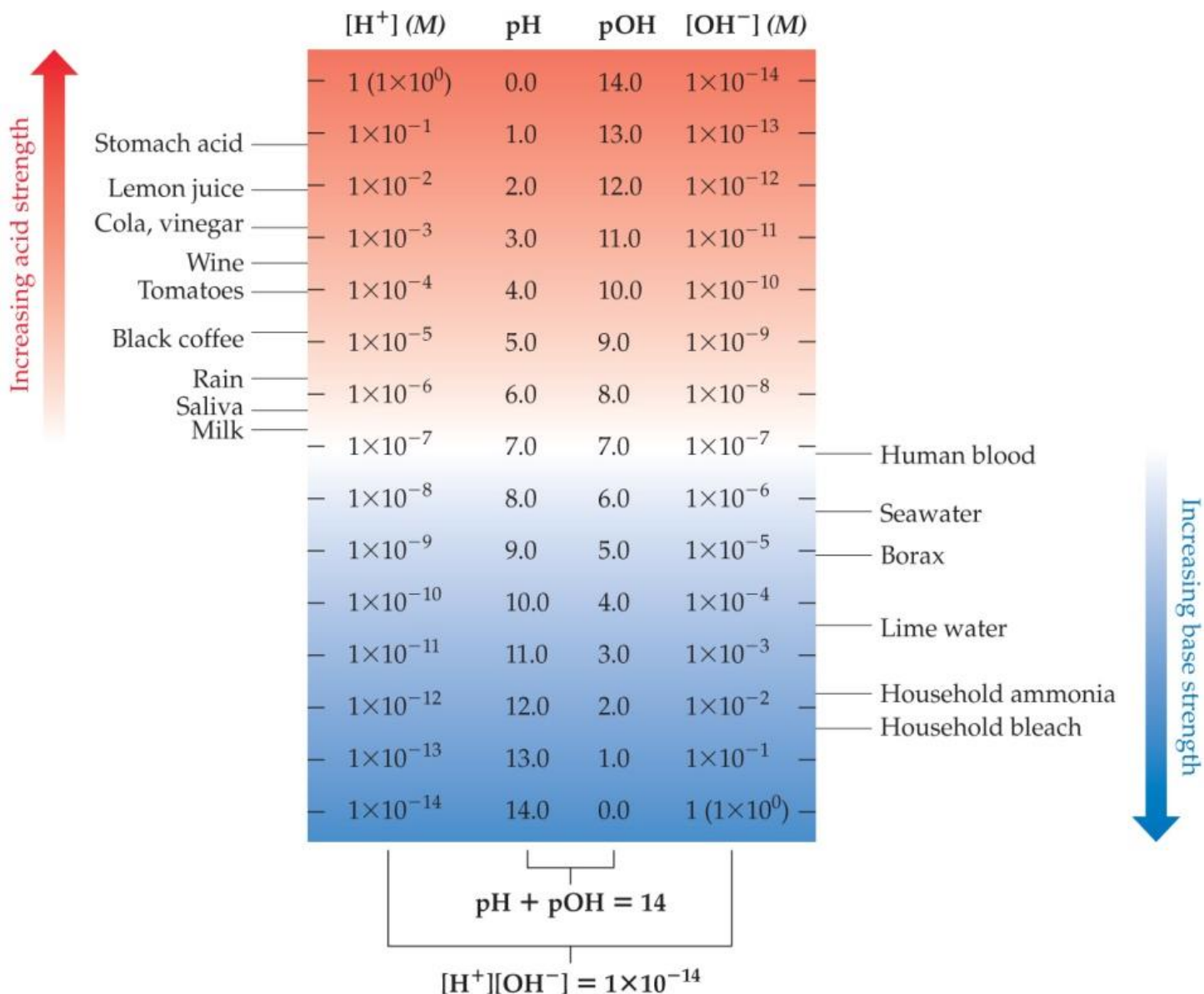
- ◆ An **acid** has a higher $[\text{H}_3\text{O}^+]$ than pure water, so its **pH < 7**.
- ◆ A **base** has a lower $[\text{H}_3\text{O}^+]$ than pure water, so its **pH > 7**.

TABLE 16.1 • Relationships among (H^+) , (OH^-) , and pH at 25 °C

Solution Type	$[\text{H}^+]$ (M)	$[\text{OH}^-]$ (M)	pH
Acidic	$>1.0 \times 10^{-7}$	$<1.0 \times 10^{-7}$	<7.00
Neutral	1.0×10^{-7}	1.0×10^{-7}	7.00
Basic	$<1.0 \times 10^{-7}$	$>1.0 \times 10^{-7}$	>7.00

pH of Acids and Bases

These are the pH values for several common substances



pOH and Other “p” Scales

- ◆ The “p” in pH tells us to take the negative logarithm in base 10 of a quantity (in this case, concentration of hydronium ions).
- ◆ Some similar examples are
 - pOH: $-\log [\text{OH}^-]$
 - $\text{p}K_w$: $-\log K_w$

pOH and Other “p” Scales

Because

$$[\text{H}_3\text{O}^+] [\text{OH}^-] = K_w = 1.0 \times 10^{-14}$$

we know that

$$-\log [\text{H}_3\text{O}^+] + -\log [\text{OH}^-] = -\log K_w = 14.00$$

or, in other words,

$$\text{pH} + \text{pOH} = \text{p}K_w = 14.00$$

EXAMPLE

A sample of freshly pressed apple juice has a pOH of 10.24. Calculate $[H^+]$.

$$pH = 14.00 - pOH$$

$$pH = 14.00 - 10.24 = 3.76$$

$$pH = -\log[H^+] = 3.76$$

$$\log[H^+] = -3.76$$

$$[H^+] = \text{antilog}(-3.76) = 10^{-3.76} = 1.7 \times 10^{-4} M$$

Measuring pH

- ◆ For less accurate measurements, one can use litmus paper or an indicator

	pH range for color change									
	0	2	4	6	8	10	12	14		
Methyl violet	Yellow		Violet							
Thymol blue	Red		Yellow		Yellow		Blue			
Methyl orange		Red		Yellow						
Methyl red			Red		Yellow					
Bromthymol blue				Yellow		Blue				
Phenolphthalein					Colorless		Pink			
Alizarin yellow R						Yellow		Red		

Measuring pH

For more accurate measurements, one uses a pH meter, which measures the voltage in the solution.



16.5 Strong Acids and Bases

	ACID		BASE	
Strong acids	HCl	↑ Base strength increasing ↓ Acid strength increasing	Cl ⁻	Negligible basicity
	H ₂ SO ₄		HSO ₄ ⁻	
	HNO ₃		NO ₃ ⁻	
	H ₃ O ⁺ (aq)		H ₂ O	
Weak acids	HSO ₄ ⁻		SO ₄ ²⁻	Weak bases
	H ₃ PO ₄		H ₂ PO ₄ ⁻	
	HF		F ⁻	
	CH ₃ COOH		CH ₃ COO ⁻	
	H ₂ CO ₃		HCO ₃ ⁻	
	H ₂ S		HS ⁻	
	H ₂ PO ₄ ⁻		HPO ₄ ²⁻	
	NH ₄ ⁺		NH ₃	
	HCO ₃ ⁻		CO ₃ ²⁻	
	HPO ₄ ²⁻		PO ₄ ³⁻	
Negligible acidity	H ₂ O		OH ⁻	Strong bases
	OH ⁻		O ²⁻	
	H ₂		H ⁻	
	CH ₄		CH ₃ ⁻	

Strong Acids

- ◆ The seven most common strong acids include
 - six monoprotic acids HCl, HBr, HI, HNO₃, HClO₃, HClO₄ and
 - one diprotic acid H₂SO₄.
- ◆ They are strong electrolytes and exist totally as ions in aqueous solution.
- ◆ For the **monoprotic strong acids**,
 $[\text{H}_3\text{O}^+] = [\text{acid}]$

Example: In a 0.20 M solution of HNO₃(aq), $[\text{H}^+] = [\text{NO}_3^-] = 0.20 \text{ M}$

Strong Bases

- ◆ Strong bases are the soluble hydroxides, which are the alkali metal and heavier alkaline earth metal hydroxides (Ca^{2+} , Sr^{2+} , and Ba^{2+}).
- ◆ These substances dissociate completely in aqueous solution.

EXAMPLE

What is the pH of

(a) a 0.040 M solution of HClO_4

Because HClO_4 is a strong acid, it is completely ionized in water, giving $[\text{H}^+] = [\text{ClO}_4^-] = 0.040 \text{ M}$

$$\text{pH} = -\log(0.040) = 1.40$$

(b) a 0.0011 M solution of $\text{Ca}(\text{OH}_2)$

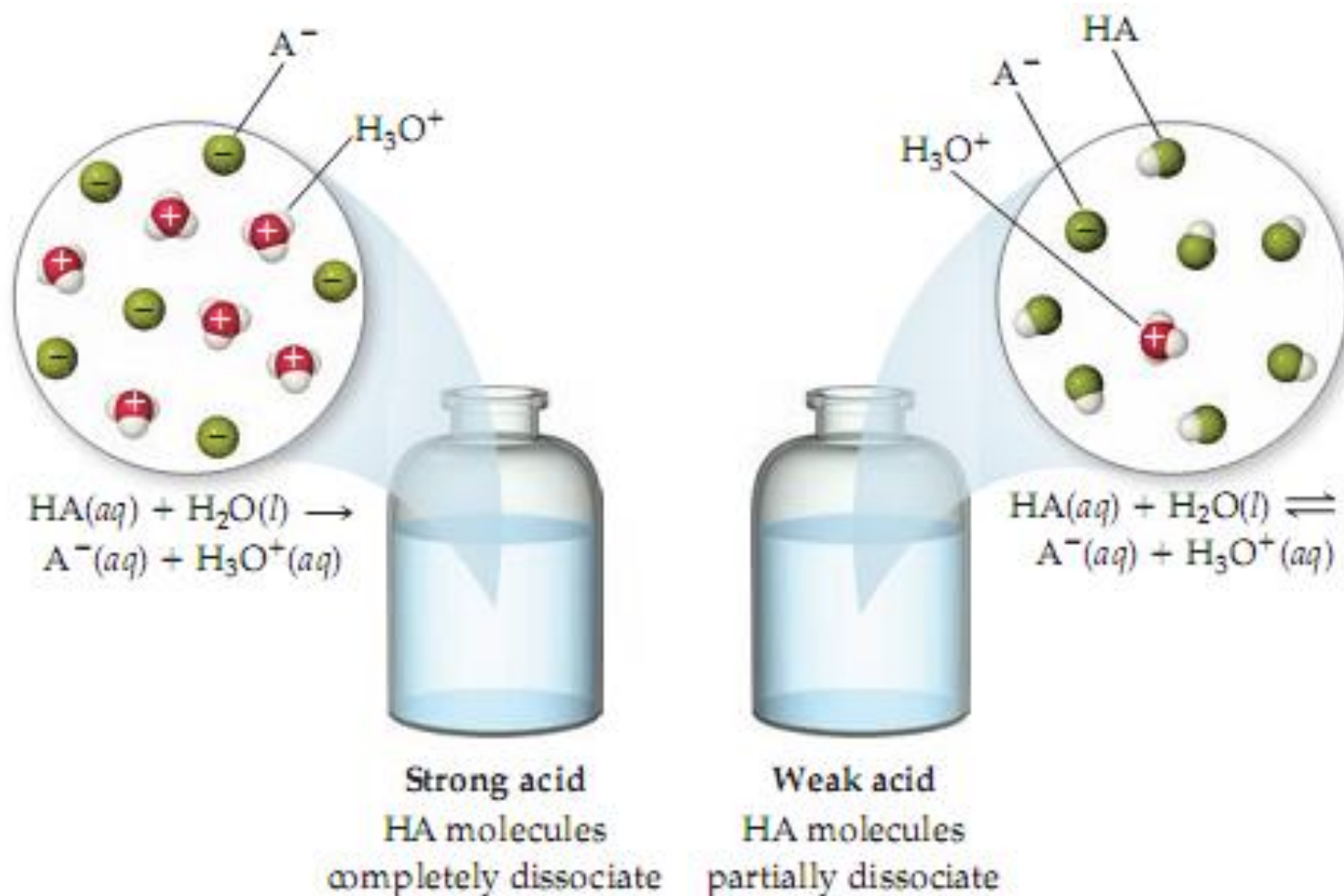
$\text{Ca}(\text{OH}_2)$ is a strong base that dissociates in water to give two OH^- ions per formula unit, giving $[\text{OH}^-] = 2 \times 0.0011 = 0.0022 \text{ M}$

$$\text{pOH} = -\log(0.0022) = 2.66$$

$$\text{pH} = 14.00 - \text{pOH} = 11.34$$

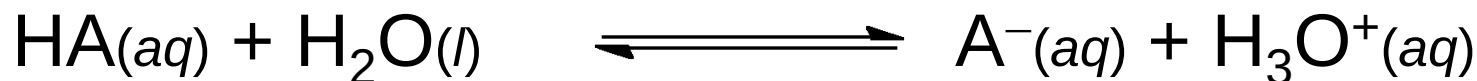
16-6 Weak Acids

- ◆ Most acidic substances are weak acids and therefore only partially ionized in aqueous solution



Dissociation Constants

- ◆ For a generalized acid dissociation,



the equilibrium expression would be

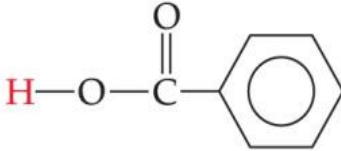
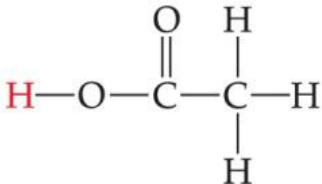
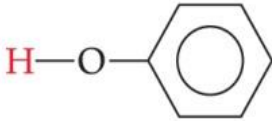
$$K_c = \frac{[\text{H}_3\text{O}^{+}] [\text{A}^{-}]}{[\text{HA}]}$$

- ◆ This equilibrium constant is called the **acid-dissociation constant**, K_a .

Dissociation Constants

The greater the value of K_a , the stronger is the acid.

TABLE 16.2 • Some Weak Acids in Water at 25 °C

Acid	Structural Formula*	Conjugate Base	K_a
Chlorous (HClO_2)	$\text{H}-\text{O}-\text{Cl}-\text{O}$	ClO_2^-	1.0×10^{-2}
Hydrofluoric (HF)	$\text{H}-\text{F}$	F^-	6.8×10^{-4}
Nitrous (HNO_2)	$\text{H}-\text{O}-\text{N}=\text{O}$	NO_2^-	4.5×10^{-4}
Benzoic ($\text{C}_6\text{H}_5\text{COOH}$)		$\text{C}_6\text{H}_5\text{COO}^-$	6.3×10^{-5}
Acetic (CH_3COOH)		CH_3COO^-	1.8×10^{-5}
Hypochlorous (HOCl)	$\text{H}-\text{O}-\text{Cl}$	OCl^-	3.0×10^{-5}
Hydrocyanic (HCN)	$\text{H}-\text{C}\equiv\text{N}$	CN^-	4.9×10^{-10}
Phenol (HOC_6H_5)		$\text{C}_6\text{H}_5\text{O}^-$	1.3×10^{-10}

*The proton that ionizes is shown in red.

Calculating K_a from the pH

The pH of a 0.10 M solution of formic acid, HCOOH, at 25°C is 2.38. Calculate K_a for formic acid at this temperature.

We know that

$$K_a = \frac{[\text{H}^+][\text{HCOO}^-]}{[\text{HCOOH}]}$$

$$\text{pH} = -\log[\text{H}^+] = 2.38$$

$$\log[\text{H}^+] = -2.38$$

$$[\text{H}^+] = 10^{-2.38} = 4.2 \times 10^{-3} \text{ M}$$

The dissociation equation: $\text{HCOOH}(aq) \rightleftharpoons \text{H}^+(aq) + \text{HCOO}^-(aq)$

Using the ICE chart

I	0.10	0	0
C			
E			

$$K_a = \frac{(4.2 \times 10^{-3})(4.2 \times 10^{-3})}{(0.10 - 4.2 \times 10^{-3})} = 1.8 \times 10^{-4}$$

Percent Ionization

◆ Percent ionization = $\frac{[\text{H}_3\text{O}^+]_{\text{eq}}}{[\text{HA}]_{\text{initial}}} \times 100$

◆ In this example,

I	0.10	0	0
C	-4.2×10^{-3}	$+4.2 \times 10^{-3}$	$+4.2 \times 10^{-3}$
E	$0.10 - 4.2 \times 10^{-3}$	4.2×10^{-3}	4.2×10^{-3}

$$[\text{H}_3\text{O}^+]_{\text{eq}} = 4.2 \times 10^{-3} \text{ M}$$

$$[\text{HCOOH}]_{\text{initial}} = 0.10 \text{ M}$$

$$\text{Percent ionization} = \frac{4.2 \times 10^{-3}}{0.10} \times 100 = 4.2\%$$

Calculating pH from K_a

1. Write the ionization equilibrium
2. Write the equilibrium-constant expression and the value for the equilibrium constant
3. Express the concentrations involved in the equilibrium reaction using ICE chart
4. Substitute the equilibrium concentrations into the equilibrium-constant expression and solve for change in concentration
5. Calculate pH

Calculating pH from K_a

Calculate the pH of a 0.30 M solution of acetic acid, $\text{HC}_2\text{H}_3\text{O}_2$, at 25°C . K_a for acetic acid at 25°C is 1.8×10^{-5} .



2. $K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 1.8 \times 10^{-5}$



I	0.30	0	0
C	-x	+x	+x
E	0.30-x	x	x

4. $K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = \frac{(x)(x)}{0.30 - x} = 1.8 \times 10^{-5}$

$$x = 2.3 \times 10^{-3} \text{ M} = [\text{H}^+]$$

5. $\text{pH} = -\log(2.3 \times 10^{-3}) = 2.64$

EXAMPLE

Calculate the pH of a 0.20 M solution of HCN. K_a for HCN is 4.9×10^{-10}



$$K_a = \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]} = 4.9 \times 10^{-10}$$



I	0.20	0	0
C	-x	+x	+x
E	0.20-x	x	x

$$K_a = \frac{(x)(x)}{0.20 - x} = 4.9 \times 10^{-10}$$

$$x = \sqrt{0.98 \times 10^{-10}} = 9.9 \times 10^{-6} \text{ M} = [\text{H}^+]$$

$$\text{pH} = -\log[\text{H}^+] = -\log(9.9 \times 10^{-6}) = 5.00$$

Polyprotic Acids

Polyprotic acids have more than one acidic proton.



- ◆ It is always easier to remove the first proton from a polyprotic acid than to remove the second. Therefore $K_{a1} > K_{a2}$
- ◆ For an acid with three ionizable protons, it is easier to remove the second proton than the third. Therefore, $K_{a2} > K_{a3}$
- ◆ If the difference between the K_a for the first dissociation and subsequent K_a values is 10^3 or more, the pH generally depends *only* on the first dissociation.

Polyprotic Acids

TABLE 16.3 • Acid-Dissociation Constants of Some Common Polyprotic Acids

Name	Formula	K_{a1}	K_{a2}	K_{a3}
Ascorbic	$\text{H}_2\text{C}_6\text{H}_6\text{O}_6$	8.0×10^{-5}	1.6×10^{-12}	
Carbonic	H_2CO_3	4.3×10^{-7}	5.6×10^{-11}	
Citric	$\text{H}_3\text{C}_6\text{H}_5\text{O}_7$	7.4×10^{-4}	1.7×10^{-5}	4.0×10^{-7}
Oxalic	$\text{H}_2\text{C}_2\text{O}_4$	5.9×10^{-2}	6.4×10^{-5}	
Phosphoric	H_3PO_4	7.5×10^{-3}	6.2×10^{-8}	4.2×10^{-13}
Sulfurous	H_2SO_3	1.7×10^{-2}	6.4×10^{-8}	
Sulfuric	H_2SO_4	Large	1.2×10^{-2}	
Tartaric	$\text{H}_2\text{C}_4\text{H}_4\text{O}_6$	1.0×10^{-3}	4.6×10^{-5}	

EXAMPLE

The solubility of CO_2 in water at 25°C and 0.1 atm is 0.0037 M . The common practice is to assume that all the dissolved CO_2 is in the form of carbonic acid (H_2CO_3), which is produced in the reaction:



(a) What is the pH of a 0.0037 M solution of H_2CO_3 ?

(b) What is the concentration of CO_3^{2-} ?

For H_2CO_3 , $K_{a1} = 4.3 \times 10^{-7}$, $K_{a2} = 5.6 \times 10^{-11}$

K_{a1} and K_{a2} differ by more than a factor of 10^3 the pH can be determined by considering only K_{a1}



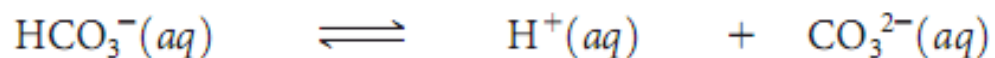
I	0.0037	0	0
C	-x	+x	+x
E	0.0037-x	x	x

$$K_{a1} = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} = \frac{(x)(x)}{0.0037 - x} = 4.3 \times 10^{-7}$$

$$x = 4.0 \times 10^{-5}\text{ M}$$

$$\text{pH} = -\log(4.0 \times 10^{-5}) = 4.40$$

EXAMPLE



I	4.0×10^{-5}	4.0×10^{-5}	0
C	-y	+y	+y
E	$4.0 \times 10^{-5} - y$	y	y

$$K_{a2} = \frac{[\text{H}^+][\text{CO}_3^{2-}]}{[\text{HCO}_3^-]} = \frac{(4.0 \times 10^{-5})(y)}{4.0 \times 10^{-5}} = 5.6 \times 10^{-11}$$

$$y = 5.6 \times 10^{-11} \text{ M} = [\text{CO}_3^{2-}]$$

16-7 Weak Bases

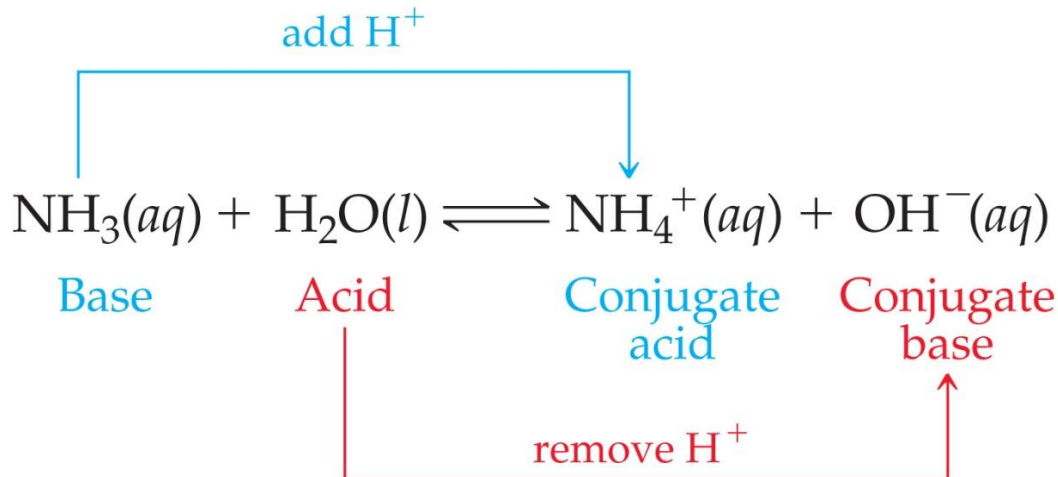
Bases react with water to produce hydroxide ion.



The equilibrium constant expression for this reaction is

$$K_b = \frac{[BH^+][OH^-]}{[B]}$$

K_b is the base-dissociation constant



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$$

K_b can be used to find $[\text{OH}^-]$ and, through it, pH.

TABLE 16.4 • Some Weak Bases in Water at 25 °C

Base	Structural Formula*	Conjugate Acid	K_b
Ammonia (NH_3)	$\begin{array}{c} \text{H}-\ddot{\text{N}}-\text{H} \\ \\ \text{H} \end{array}$	NH_4^+	1.8×10^{-5}
Pyridine ($\text{C}_5\text{H}_5\text{N}$)		$\text{C}_5\text{H}_5\text{NH}^+$	1.7×10^{-9}
Hydroxylamine (HONH_2)	$\begin{array}{c} \text{H}-\ddot{\text{N}}-\ddot{\text{O}}\text{H} \\ \\ \text{H} \end{array}$	HONH_3^+	1.1×10^{-8}
Methylamine (CH_3NH_2)	$\begin{array}{c} \text{H}-\ddot{\text{N}}-\text{CH}_3 \\ \\ \text{H} \end{array}$	CH_3NH_3^+	4.4×10^{-4}
Hydrosulfide ion (HS^-)	$[\text{H}-\ddot{\text{S}}:]^-$	H_2S	1.8×10^{-7}
Carbonate ion (CO_3^{2-})	$\left[\begin{array}{c} \ddot{\text{O}}: \\ \\ \text{C} \\ / \quad \backslash \\ \ddot{\text{O}}: \quad \ddot{\text{O}}: \end{array} \right]^{2-}$	HCO_3^-	1.8×10^{-4}
Hypochlorite ion (ClO^-)	$[:\ddot{\text{Cl}}-\ddot{\text{O}}:]^-$	HClO	3.3×10^{-7}

*The atom that accepts the proton is shown in blue.

EXAMPLE

What is the pH of a 0.15 M solution of NH_3 ? For NH_3 , $K_b = 1.8 \times 10^{-5}$



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]} = 1.8 \times 10^{-5}$$



I	0.15		0	0
C	-x		+x	+x
E	0.15-x		x	x

$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]} = \frac{(x)(x)}{0.15 - x} = 1.8 \times 10^{-5}$$

$$x = [\text{NH}_4^+] = [\text{OH}^-] = \sqrt{2.7 \times 10^{-6}} = 1.6 \times 10^{-3} M$$

$$\text{pOH} = -\log(1.6 \times 10^{-3}) = 2.80$$

$$\text{pH} = 14.00 - 2.80 = 11.20$$

16-8 Relationship Between K_a and K_b

The product of the acid-dissociation constant for an acid and the base-dissociation constant for its conjugate base equals the ion-product constant for water:

$$K_a \times K_b = K_w$$
$$pK_a + pK_b = pK_w = 14 \text{ at } 25^\circ\text{C}$$



$$K_a = \frac{[\text{NH}_3][\text{H}^+]}{[\text{NH}_4^+]}$$



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$$



$$K_a \times K_b = \left(\frac{[\text{NH}_3][\text{H}^+]}{[\text{NH}_4^+]} \right) \left(\frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]} \right)$$
$$= [\text{H}^+][\text{OH}^-] = K_w$$

Table 16.5 Some Conjugate Acid–Base Pairs

Acid	K_a	Base	K_b
HNO ₃	(Strong acid)	NO ₃ [−]	(Negligible basicity)
HF	6.8×10^{-4}	F [−]	1.5×10^{-11}
CH ₃ COOH	1.8×10^{-5}	CH ₃ COO [−]	5.6×10^{-10}
H ₂ CO ₃	4.3×10^{-7}	HCO ₃ [−]	2.3×10^{-8}
NH ₄ ⁺	5.6×10^{-10}	NH ₃	1.8×10^{-5}
HCO ₃ [−]	5.6×10^{-11}	CO ₃ ^{2−}	1.8×10^{-4}
OH [−]	(Negligible acidity)	O ^{2−}	(Strong base)

16-9 Acid–Base Properties of Salt Solutions

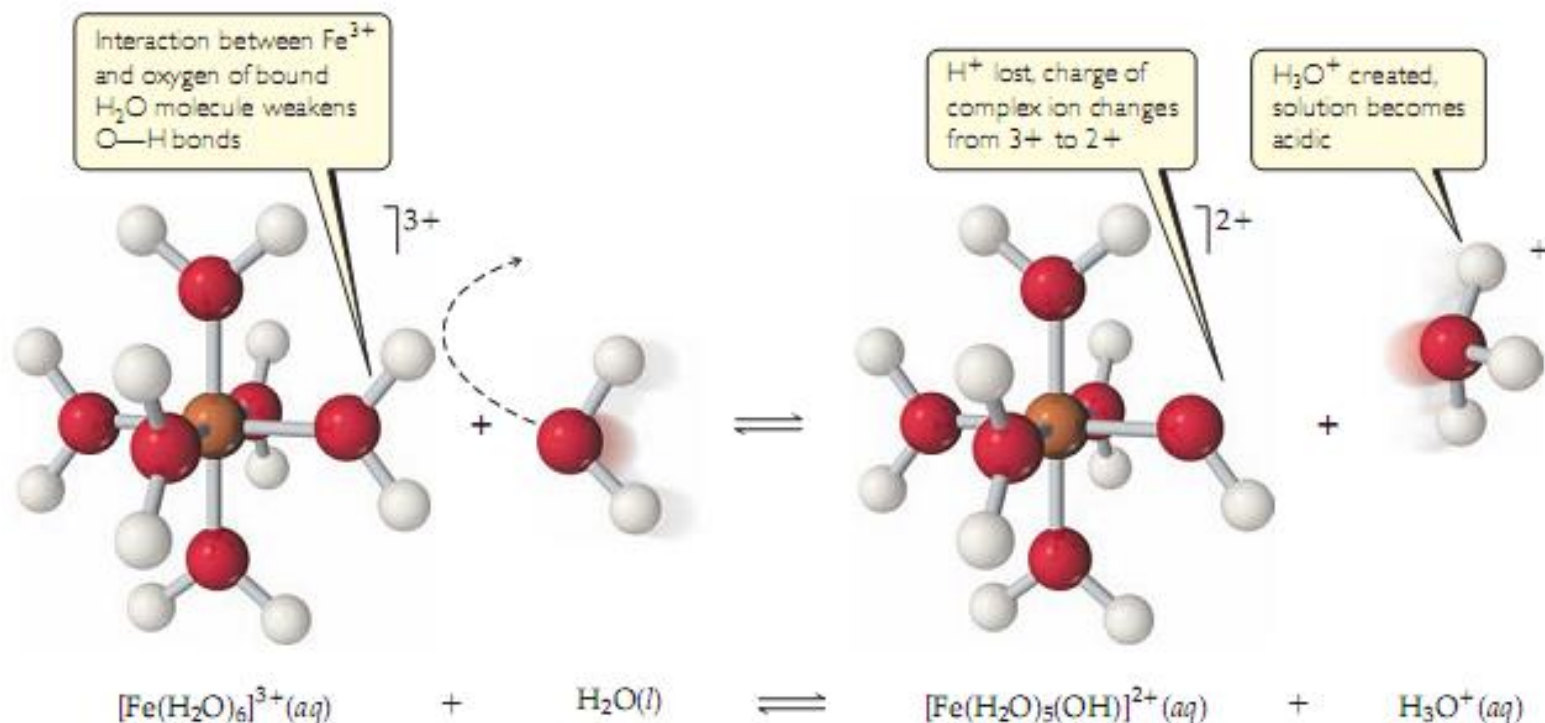
- ◆ Anions can be considered the conjugate bases of acids.
 - As such, they can react with water in a **hydrolysis reaction** to form OH^- and the conjugate acid:



- ◆ Polyatomic cations containing one or more protons can be considered the conjugate acids of weak bases.
 - As such, they can **donate a proton to water**, producing hydronium ions and thereby lowering the pH



Reactions of Cations with Water



- ◆ Attraction between nonbonding electrons on oxygen and the metal causes a shift of the electron density in water.
- ◆ This makes the O—H bond more polar and the water more acidic.

Reactions of Cations with Water

TABLE 16.6 • Acid-Dissociation Constants for Metal Cations in Aqueous Solution at 25 °C

Cation	K_a
Fe^{2+}	3.2×10^{-10}
Zn^{2+}	2.5×10^{-10}
Ni^{2+}	2.5×10^{-11}
Fe^{3+}	6.3×10^{-3}
Cr^{3+}	1.6×10^{-4}
Al^{3+}	1.4×10^{-5}

- ◆ Most metal cations that are hydrated in solution also lower the pH of the solution
- ◆ Attraction between nonbonding electrons on oxygen and the metal causes a shift of the electron density in water.
- ◆ This makes the O–H bond more polar and the water more acidic.
- ◆ Greater charge and smaller size make a cation more acidic

Combined Effect of Cation and Anion in Solution

1. An anion that is the conjugate base of a strong acid will not affect the pH.
2. An anion that is the conjugate base of a weak acid will increase the pH.
3. A cation that is the conjugate acid of a weak base will decrease the pH.
4. Cations of the strong Arrhenius bases will not affect the pH.
5. Other metal ions will cause a decrease in pH.
6. When a solution contains both a weak acid and a weak base, the affect on pH depends on the K_a and K_b values.

EXAMPLE

Determine whether aqueous solutions of each of these salts are acidic, basic, or neutral:

(a) $\text{Ba}(\text{CH}_3\text{COO})_2$ **Basic**

(b) NH_4Cl **Acidic**

(c) $\text{CH}_3\text{NH}_3\text{Br}$, **Acidic**

(d) KNO_3 , **Neutral**

(e) $\text{Al}(\text{ClO}_4)_3$ **Acidic**

16-10 Acid–Base Behavior and Chemical Structure

Factors Affecting Acid Strength

- ◆ The more polar the H–X bond the more acidic the compound
- ◆ The weaker the H–X bond, the more acidic the compound.
- ◆ The greater the stability of the conjugate base, the stronger the acid.

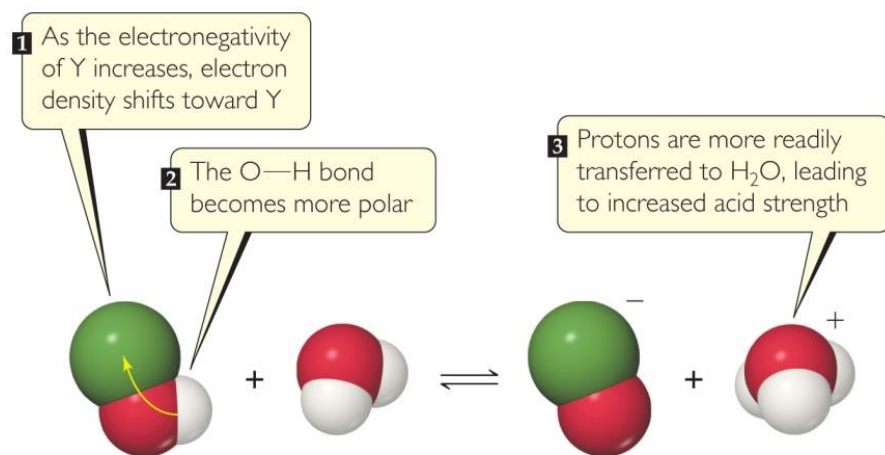
4A	5A	6A	7A	
CH₄ Neither acid nor base	NH₃ Weak base $K_b = 1.8 \times 10^{-5}$	H₂O	HF Weak acid $K_a = 6.8 \times 10^{-4}$	Increasing acid strength ↓
SiH₄ Neither acid nor base	PH₃ Very weak base $K_b = 4 \times 10^{-28}$	H₂S Weak acid $K_a = 9.5 \times 10^{-8}$	HCl Strong acid	
		H₂Se Weak acid $K_a = 1.3 \times 10^{-4}$	HBr Strong acid	
Increasing acid strength →				

Binary Acids

- ◆ As the element A increases in size → The strength of an H – A bond tends to decrease → The acidity **increases down a group**
 - HCl is a stronger acid than HF
 - H₂S is a stronger acid than H₂O
- ◆ Bond polarity is the major factor determining acidity: acidity increases as the electronegativity of the element A increases
 - C – H bond is essentially nonpolar, CH₄ shows no tendency to form H⁺ and CH₃⁻ ions.
 - N – H bond is polar, but NH₃ has a lone pair of electrons on the nitrogen atom that dominates its chemistry, so NH₃ acts as a base rather than an acid

Oxyacids

In oxyacids, in which an –OH is bonded to another atom, Y, the more electronegative Y is, the more acidic the acid.



Substance	Y—OH	Electronegativity of Y	Dissociation constant
Hypochlorous acid	Cl—OH	3.0	$K_a = 3.0 \times 10^{-8}$
Hypobromous acid	Br—OH	2.8	$K_a = 2.5 \times 10^{-9}$
Hypoiodous acid	I—OH	2.5	$K_a = 2.3 \times 10^{-11}$
Water	H—OH	2.1	$K_w = 1.0 \times 10^{-14}$

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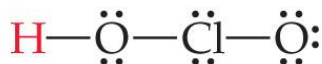
Oxyacids

Hypochlorous



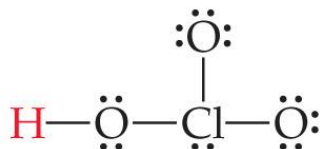
$$K_a = 3.0 \times 10^{-8}$$

Chlorous



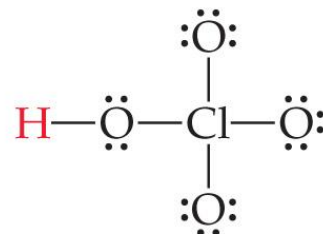
$$K_a = 1.1 \times 10^{-2}$$

Chloric



Strong acid

Perchloric



Strong acid

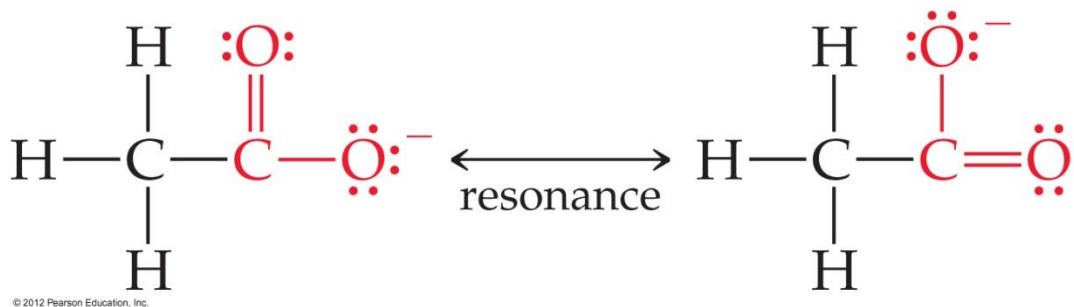
Increasing acid strength



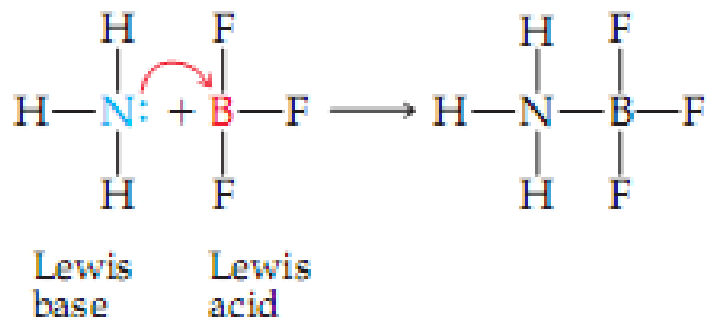
For a series of oxyacids, acidity increases with the number of oxygens.

Carboxylic Acids

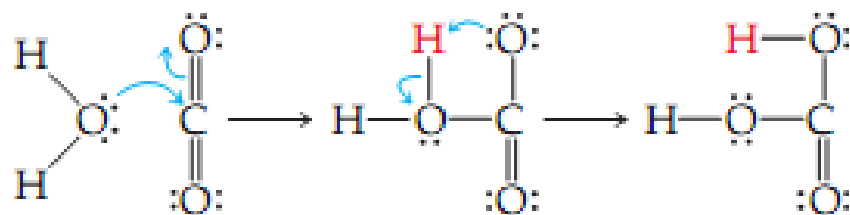
Resonance in the conjugate bases of carboxylic acids stabilizes the base and makes the conjugate acid more acidic.



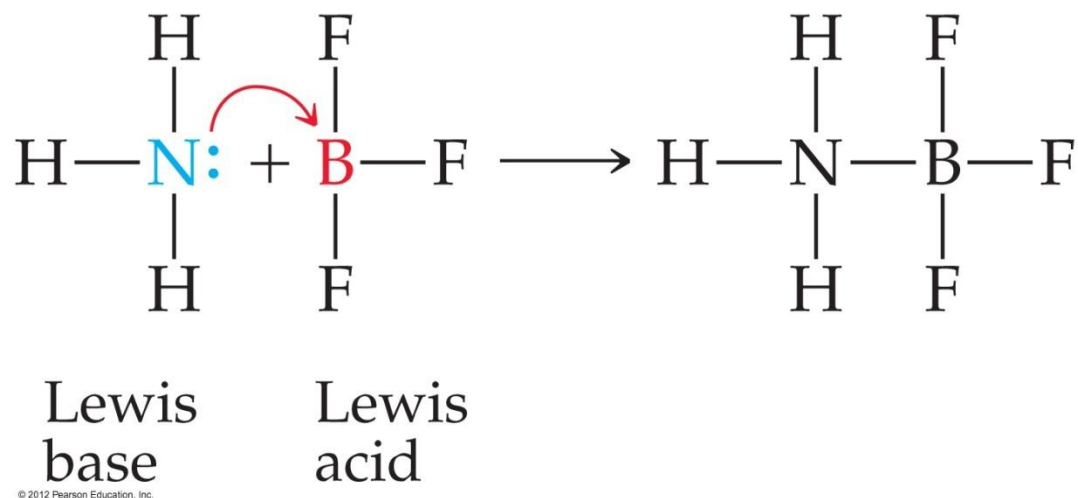
16.11 | Lewis Acids and Bases



- ◆ **Lewis acids** are defined as electron-pair acceptors.
- ◆ Atoms with an empty valence orbital can be Lewis acids.



Lewis Bases



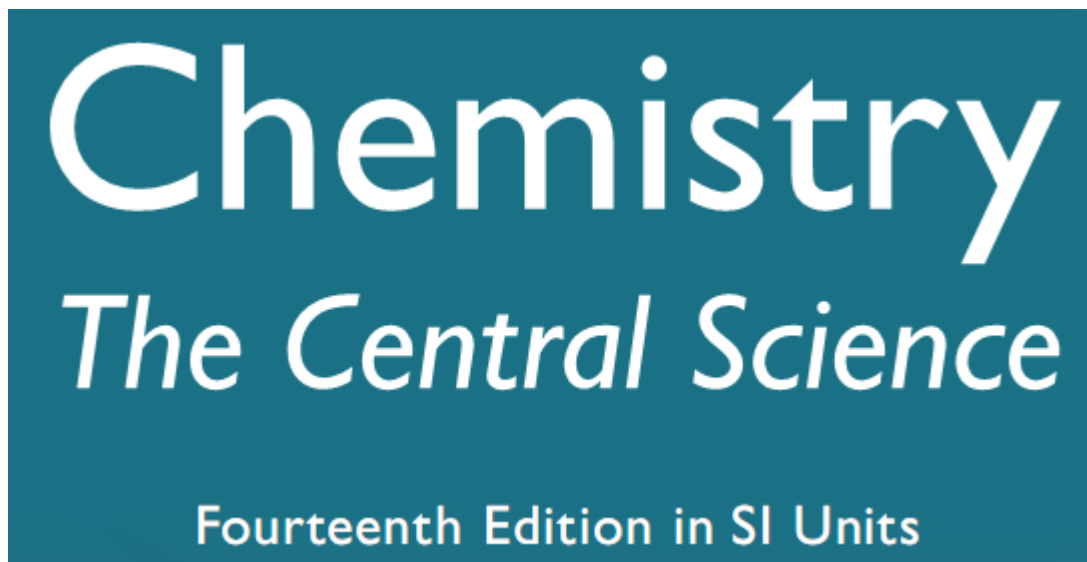
- ◆ **Lewis bases** are defined as electron-pair donors.
- ◆ Anything that could be a Brønsted–Lowry base is a Lewis base.
- ◆ Lewis bases can interact with things other than protons, however.

Homeworks

16.119

16.126

Due: August 14th 2019



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It's a quick reminder that Group 1 and Group 2 will present their research topics on Wednesday August 14th 2019